

NIKHIL GADDAM

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EDUCATION

Masters in Computer Science, Virginia Tech, USA *August 2023 - December 2024*
Bachelors of Technology in CE, Indian Institute of Technology, Guwahati *July 2017 - July 2021*

EXPERIENCE

Strategy | Software Engineer 2 - Infrastructure **January 2025 - Present**
Formerly MicroStrategy *Tysons Corner, Virginia*

- Cut tenant migration time by **67% (6 hrs to 2 hrs)** by building an instance-to-Kubernetes migration engine on **Temporal** workflows that auto-provision tenant networks, clusters, and environments from backups, with auto-computed cluster sizing and automated network onboarding into existing customer VPCs
- Reduced cloud spend by **\$10M annually (25% efficiency gain)** by building adaptive **autoscaling** for services with **KEDA** and a **Prometheus**-driven metrics pipeline tuned to workload spikes
- Designed a distributed multi-cloud Kubernetes platform across AWS, GCP, and Azure with custom **controllers in Go** for fleet-wide cluster provisioning, delivering zero-downtime infrastructure upgrades via **Istio** service mesh
- Architected a cloud-native **agentic AI platform** for Kubernetes operations with model-agnostic **LLM routing** and **multi-agent orchestration** (planning, **RAG**-based context retrieval, and tool-calling agents), adding guardrails, human-in-the-loop approvals to automate incident triage, alerting, cost optimization, and infra audits
- Architected the multi-cloud persistent storage layer for stateful Kubernetes workloads (**ClickHouse**, **Redis**) on GCP, AWS, and Azure: ran zero-loss data migrations, re-tiered volumes (zonal to regional) for availability and throughput, and eliminated recurring storage incidents via automated guardrails

Strategy | Software Engineer Intern **May 2024 - August 2024**
Formerly MicroStrategy *Tysons Corner, Virginia*

- Designed a high-throughput search auto-completion engine using **RAG + LangChain**, optimizing embedding retrieval and caching paths; improved suggestion accuracy and cut end-to-end query latency by 60 % across 10K+ daily users.
- Built a predictive SQL-caching layer that analyzes historical workloads to pre-fetch frequent queries, reducing search engine load by 50% and improving dashboard render times under peak enterprise traffic.

Sprinklr | Software Engineer **July 2021 – August 2023**
Unified AI Customer Experience Management SaaS *Delhi, India*

- Developed core backend components for Sprinklr's Conversational AI platform, leveraging large language models and unsupervised learning to automate intent discovery and improve natural language understanding (NLU).
- Optimized the no-code business process automation platform, improving workflow execution efficiency and cutting **first-response times by 33% for Uber and Microsoft** across high-volume customer operations.
- Designed REST APIs for the Guided Workflows product, adopted by 15+ teams to improve cross-team automation

PROJECTS

Archon | Python [GitHub](#)
• Open-source contributor to Archon, a modular agentic Operating System for building autonomous multi-agents.

Agent Browser for AI Agents | TypeScript | Python [GitHub](#)
• Contributed to an open-source headless browser automation CLI for AI agents built for LLM optimized Navigation

RESEARCH

Identifying Parking Demand Hotspots and Analyzing Rate Surges in Seattle *November 2024*
Oral Presentation at Urban Transitions 2024 *Sitges, Spain*

ACHIEVEMENTS

Joint Entrance Examination: Secured an AIR 122 (top 0.08%) among 1.18 million candidates

TECHNICAL SKILLS

Languages: Go, Java, C++, Python, TypeScript, JavaScript, Bash, SQL

Technologies/Frameworks: Kubernetes, controller-runtime, KEDA, Prometheus, Helm, ArgoCD, Istio, Docker, Terraform, OpenShift, ClickHouse, Redis, Kafka, Apache Spark, gRPC, Elasticsearch, RESTful APIs, GraphQL, WebSockets, Pandas, NumPy, Spring Boot, LangChain

Certifications: Certified Kubernetes Application Developer (CKAD)